TEJAS N

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PROFESSIONAL SUMMARY

Third-year B.Tech Computer Science student at MIT Bengaluru specializing in AI/ML engineering and full-stack web development. Hands-on experience building production-ready AI applications including RAG systems, LLM-powered tools, and data analytics platforms using Python, React.js, LangChain-Graph and HuggingFace.

EDUCATION

B.Tech, Computer Science and Engineering August 2024 – May 2028
Manipal Institute of Technology, Bengaluru | CGPA: 7.4/10 (up to 4th Semester)
Relevant Coursework: DSA, Database Management Systems, Machine Learning, OOPS, Operating Systems, Computer Networks

TECHNICAL SKILLS

Languages: Python, JavaScript, C, HTML, CSS
Frameworks: React.js, Node.js, Streamlit
AI / ML: scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, FAISS, HuggingFace Transformers, LangChain, LangGraph
Tools: Git, GitHub, Firebase, Docker, N8N, REST APIs, VS Code, CI/CD
Concepts: RAG, NLP, Prompt Engineering, Semantic Search, Transformer Architecture, Model Training, Data Pipelines, Full-Stack Development

PROJECTS

SATQ AI - Geospatial Foundation Platform Sep 2026 onwards - SIH Hackathon
React.js · FastAPI · Pytorch · Gemini 2.5 Flash · Grounding Dino · Meta SAM 2 · PostGIS · OpenCV · Leaflet.js

- Built an agentic platform using Gemini 2.5 Flash as an AI orchestrator to parse natural language queries, cutting spatial analysis time from 4+ hours to <10 seconds.
- Engineered a dynamic ROI engine using Leaflet.js and OpenCV to stream 512x512 tile crops, reducing GPU compute overhead by over 70%.
- Integrated a multimodal AI pipeline (Grounding DINO, SAM 2, Prithvi-EO, Siam-Swin ViT) for zero-shot segmentation, change detection, and auditable GeoJSON export.

AI-Powered RAG Question Answering System Jan 2025 – Mar 2025
Python · FAISS · HuggingFace Flan-T5 · LangChain · Streamlit · NLP · Semantic Search

- Developed a RAG pipeline enabling context-aware Q&A over custom document corpora using FAISS vector retrieval, HuggingFace Flan-T5, and LangChain; achieved sub-500ms query latency.
- Implemented document chunking and semantic embedding strategies, reducing document lookup time by ~70% compared to manual search across 100+ page corpora.
- Designed a Streamlit interface supporting dynamic document ingestion and real-time query processing, making the system accessible to non-technical users.

CDS — AI-Powered Crack Detection System | Hackverse 2.0 — IBM | Aug 2026
React.js · FastAPI · OpenCV · OpenRouter · Google Gemini · IBM Granite · SQLite

- Built an AI-powered infrastructure inspection platform supporting **image, video, and live-camera** crack detection with automated severity assessment and reporting.
- Designed a **multi-provider AI pipeline** using OpenRouter, Gemini, and OpenCV fallback for resilient defect detection and structured bounding-box outputs.
- ***Selected for the final round among 90 teams*** *at Hackverse 2.0, an IBM-judged hackathon.*

CERTIFICATIONS

IBM — Fundamentals of AI Agents Using RAG and LangChain
Meta — Front-End Development
Microsoft — Data Analysis and Visualization in Python
BCG X — GenAI Job Simulation | AI-powered financial chatbot development